PLL Characterisation (HEF4046B): Lab Report Outline

The manual says there is no universal structure, and the marks go to technique, analysis and handling of imperfections. This is an outline only. The manual forbids having AI write substantial parts of the report, so the prose should be yours.

Title and Abstract (~150–200 words, written last)

One sentence each, with numbers:

1. What was studied (4046 PC1/PC2, VCO, closed loop, plus the applications you did).
2. Method (automated PicoScope/Python sweeps, Arduino phase supply).
3. Headline results with uncertainties: K_D(PC1) vs V_{DD}/π, K_D(PC2) vs $V_{DD}/4\pi$, K_O, lock range vs prediction, ζ and ω_n, longest gap survived, etc.
4. One-line conclusion (e.g. which comparator suits which application).

1. Introduction (~0.5 page)

- Purpose of a PLL, and the three blocks (comparator, filter, VCO).
- Aims: characterise the blocks independently, then close the loop, then apply it.
- Scope: say plainly which Section 7 applications you attempted, and why only those.

2. Theory (~1.5–2 pages)

Derive only what you use later.

2.1 Loop model

- $\omega_o = \omega_c + K_O(V_e - V_{e0})$, $V_e = K_D \cdot \varphi_e$
- $G(s) = K_v/(s(1 + s\tau))$, $K_v = K_O K_D$, $\tau = R_f C_f$
- $H(s) = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$, $\omega_n = \sqrt{(K_v/\tau)}$, $\zeta = 1/(2\sqrt{(K_v\tau)})$
- Phase margin: $PM = 90° - \arctan(\omega_c\tau)$
- Justify the small-signal assumption: $V_e \approx$ constant over a VCO cycle, which needs $\omega_n \ll \omega_c$.

2.2 Comparators

- PC1: $\bar{V} = V_{DD}\cdot\varphi/\pi$, $K_D = V_{DD}/\pi$ (linear range 0 to π).

- PC2: V̄ = V_DD/2 + V_DD·φ/(4π), K_D = V_DD/(4π) (linear range ±2π).
- State-diagram argument for PC2 hold behaviour and zero static phase error.

2.3 VCO

- Current mirror: I_c = I_in.
- f = I_c/(2C·ΔV_pp), with I_in = I_1 + I_2, I_1 ≈ (V_in − V_GS)/R1, I_2 ≈ V_2/R2.
- Predictions: slope ∝ 1/(R1·C), f_min ∝ 1/(R2·C).
- Qin as a source follower, with the MOSFET equations if you did 7.4.

2.4 Lock and capture

- PC1: Δω_L = K_O·V_DD/2, with the capture-range approximation Δω_C ≈ Δω_L/√(1 + Δω_C²τ²).
- PC2: capture ≈ lock range.
- Static phase error: φ_e = π/2 + Δω/K_v (PC1), φ_e → 0 (PC2).

2.5 Application-specific theory

One short subsection per application done.

- Multiplier: divider by N changes the gain to K_v/N.
- FM / transient: step response, M_p, t_s, f_−3dB.
- Clock regeneration: gap drift Δφ = K_O·ΔV_e·τ(1 − e^(−t_g/τ)).

3. Experimental method (~1.5 pages)

- **3.1 Apparatus:** a block diagram figure; list the components with measured values, V_DD actually measured, and PicoScope settings (range, probe, sample rate, coupling).
- **3.2 Automation:** Python/Arduino flow in two or three lines, not code listings.
- **3.3 Measurement procedures:** short subsection per Section 6/7 experiment.
- **3.4 Data processing:** edge interpolation, averaging over an integer number of periods, fitting method, and how uncertainties were propagated.
- **3.5 Error sources identified up front:** Arduino jitter, quantisation of the 8-bit scope, component tolerance and stray capacitance, supply drift, loading by the probe.

4. Results and analysis (the bulk, ~4–5 pages)

One subsection per measurement, in the manual's order. In each, state the prediction, show the data, give the fitted value with uncertainty, and compare.

§	Figure/table	Key quantity
4.1 Phase shifter	commanded vs measured φ	step size, jitter
4.2 PC1	waveforms; V̄ vs φ with fit	K_D vs V_DD/π
4.3 PC2	waveforms; V̄ vs φ with fit	K_D vs V_DD/4π
4.4 VCO	f vs V_in for all R1/R2; scaling collapse	f_min, slope, K_O, ΔV_pp, C_s
4.5 VCO currents	I_c vs I_in	mirror ratio
4.6 Closed loop	φ_e vs f_in; lock/capture ranges; capacitor waveforms; noise data	K_v, Δω_L, Δω_C, σ_φ vs SNR
4.7+ Applications	one subsection each	ζ, ω_n, t_s, f_−3dB, gap tolerance, etc.

Include in 4.4 the comparison of the two days of VCO data (z-scores or reduced χ^2), and state which you used and why.

5. Discussion (~1–1.5 pages)

- **Consistency checks** (this is where marks are won):
  - K_O from the sweeps vs K_O from the FM step.
  - K_v from the φ_e slope vs K_O·K_D.
  - Lock range predicted vs measured.
  - ΔV_pp from the fit vs ΔV_pp from the scope.
- **Where the model fails and why.** Examples are nonlinearity at the VCO range edges, finite output impedance, and the PC1 harmonic-locking behaviour.
- **PC1 vs PC2 comparison:** noise sensitivity, capture, static phase error, recovery from interference, and which is better for each application.
- **Limitations and sources of systematic error**, with an estimate of their size.
- **Possible improvements** and one or two extension ideas.

6. Conclusion (~0.25 page)

A short list of the main quantitative results, each with its uncertainty, and the take-home message about comparator choice.

References

Follow the manual's reference list and add any you used:

- Cavendish E1/E2 class manual, Phase-Locked Loops (Irvine, 2026).
- NXP, HEF4046B data sheet; 74HC4046 data sheet.
- Horowitz & Hill, *The Art of Electronics* (CUP).
- Grebene, *Bipolar and MOS Analog Integrated Circuit Design* (Wiley).
- Sze, *Semiconductor Devices: Physics and Technology* (Wiley).
- PicoScope 2000 Series documentation; Arduino/ATmega328P reference.
- Gardner, *Phaselock Techniques* (Wiley), if you cite capture-range results.

Appendices (optional)

- Extra derivations (e.g. the −3 dB bandwidth formula).
- Tables of raw fitted parameters for every (R1, R2, C).
- Circuit diagrams and wiring notes.
- Code is not required, but keep it saved.

Practical notes

- Put uncertainties on every plotted point, and give residual plots for the key fits.
- Give figure captions that state the conclusion, not just the contents.
- Keep the total length proportionate. A deep treatment of Section 6 plus a few well-analysed applications beats a shallow tour of everything.
- Say explicitly when a result disagrees with prediction and what you did to investigate.